

Who Discovers the Dollar?

Banking Rails and Arbitrage in Stablecoin Markets

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Abstract

Why does the same order-flow shock have such different price impact over the trading day? This paper shows that the answer is effective arbitrage capacity, the speed at which arbitrage closes a dislocation, and that access to banking rails governs it. Arbitrageurs facing transfer frictions and convex balance-sheet costs leave the local stablecoin premium untouched inside a band and trade against it outside. In minute-level data on USDT against the Turkish lira from 2024 to 2026, the premium does not mean-revert inside a band of roughly 30 bp; above the band, it reverts three to five times faster when Istanbul banks are open, also with bands estimated by banking state. Large trades lean against dislocations while small and medium trades move with them, and the large-trade response rises with bank opening by a factor statistically indistinguishable from the price-side response. On weekday public holidays both stay at closed-bank levels. In Mexico, where the central bank’s payment system runs around the clock, arbitrage retains most of its speed at night. Corrections are detectable in the stablecoin price but not in the interbank rate.

Keywords: stablecoins, limits to arbitrage, banking rails, price discovery, order flow, threshold dynamics

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1 Introduction

A dollar stablecoin is the same claim everywhere, yet its price in local currency drifts away from the dollar for hours and days. In Turkey, one USDT bought on Binance cost more than 100 bp above the interbank dollar in the first half of 2024 and traded at a discount of 10–25 bp in 2026. Within the day, a sustained order-flow imbalance of a given size moves the premium about five times as much at night as during Istanbul business hours. Why does the same order-flow shock have such different price impact over the trading day?

The answer developed here is that the binding state variable is *effective arbitrage capacity*, the speed at which arbitrage closes a dislocation, and that access to the banking system governs it. Arbitrage between a local stablecoin market and the dollar runs through on- and off-ramps, banking rails and exchange withdrawals. These impose a per-unit cost c , and finite balance sheets make large positions increasingly costly, with capacity K governing that convexity. An arbitrageur who chooses a flow a to maximize $ab - c|a| - a^2/(2K)$, where b is the log premium of the local market over the dollar, follows the soft-threshold rule $a^*(b) = K \operatorname{sign}(b) (|b| - c)_+$. With end-user order flow q_t moving the premium with impact β and arbitrage flow moving it back with impact λ , the premium follows the threshold diffusion

$$db_t = \beta q_t dt - \kappa \operatorname{sign}(b_t) (|b_t| - c)_+ dt + \sigma dW_t, \quad \kappa \equiv \lambda K.$$

Inside the band $[-c, c]$ the premium is a martingale; outside it the excess decays at rate κ . With persistent order flow the premium settles at $b^* = c + \beta q/\kappa$, so the long-run price impact of order flow is β/κ , while steady-state arbitrage volume, $\beta q/\lambda$, does not depend on capacity: trading volume measures demand, and the premium measures capacity. A local information shock of size δ reaches the global price in proportion $\omega(1 - c/\delta)_+(1 - e^{-\kappa h})$ by horizon h , so the friction sets how much local information is transmitted and capacity sets how fast.

The mechanism is identified from three sides, using one-minute Binance data on USDT against the Turkish lira and one-minute interbank USD/TRY quotes from January 2024 to August 2026 (Figure 1).

Price evidence. The premium shows no mean reversion inside a band of median width 32 bp and reverts outside it; linear error correction is rejected. Above the band, the premium reverts three to five times faster when Istanbul banks are open; below the band the difference has the same sign and is estimated less precisely. Bands estimated separately for open and closed hours are wider at night on average, and the speed difference holds with them, so banking hours raise effective capacity over and above any effect on the friction. The difference is as large when the interbank market is actively quoted. In Mexico, where the

central bank’s settlement system runs around the clock, the same construction shows a much smaller day–night difference: arbitrage at night retains nearly two thirds of its daytime speed in Mexico and about one twentieth in Turkey, and the difference in open-to-closed ratios is significant.

Flow evidence. Trade-level data reveal two sharply different flow patterns corresponding to the two forces in the model. Small and medium trades exhibit demand-like flow: they move with the dislocation. Large trades exhibit arbitrage-like flow: in the five and thirty minutes after the premium leaves the band they trade toward it, in proportion to the distance outside, and they do not respond to the premium inside the band. The large-trade response rises with bank opening by a factor of 4.6, the price-side speed by a factor of 3.3, and a cross-equation test does not reject that the two ratios are equal. Prices and trades are thus consistent with a common shift in effective capacity. On weekday public holidays, when the clock says business hours but banks are closed, neither response rises.

Price-discovery evidence. Corrections are statistically detectable in the stablecoin price but not in the interbank rate, which responds neither to the premium nor to stablecoin order flow. On 19 March 2025 the interbank rate jumped 11% within minutes while USDT/TRY rose 6%, opening a 660 bp discount that closed within 75 minutes. The dollar’s lira price is discovered in the interbank market, and effective capacity sets how closely the stablecoin market follows it.

These estimates allow a counterfactual: if effective capacity stayed at its open-hours level around the clock, holding order flow and frictions fixed, the share of time the premium spends more than 50 bp outside its band would fall by 19%, and the average wedge paid on small and medium trades by 12%. Settlement infrastructure that runs on the clock of the crypto market, rather than of the banking system, would bring the local stablecoin price materially closer to the dollar.

Related literature. The paper sits at the intersection of four literatures.

Cross-market crypto arbitrage. [Makarov and Schoar \(2020\)](#) document large and persistent cross-country deviations in cryptocurrency prices, link them to capital controls, and show that signed volume co-moves with spreads. [Hautsch et al. \(2024\)](#) show that settlement latency limits cross-exchange arbitrage, and [Griffin and Shams \(2020\)](#) study the interaction of stablecoin issuance and crypto prices. [Hu and Zhang \(2023\)](#) model how arbitrageurs endogenously connect exchanges and show that star-shaped arbitrage networks can arise in equilibrium. [Shao and Rajapaksa \(2026\)](#) use ten-second order books and trades to study executable dislocations across BTC/USD, BTC/USDT and USDT/USD routes, finite-depth capacity and the adjustment of each leg; their routes are crypto-to-crypto, whereas the fric-

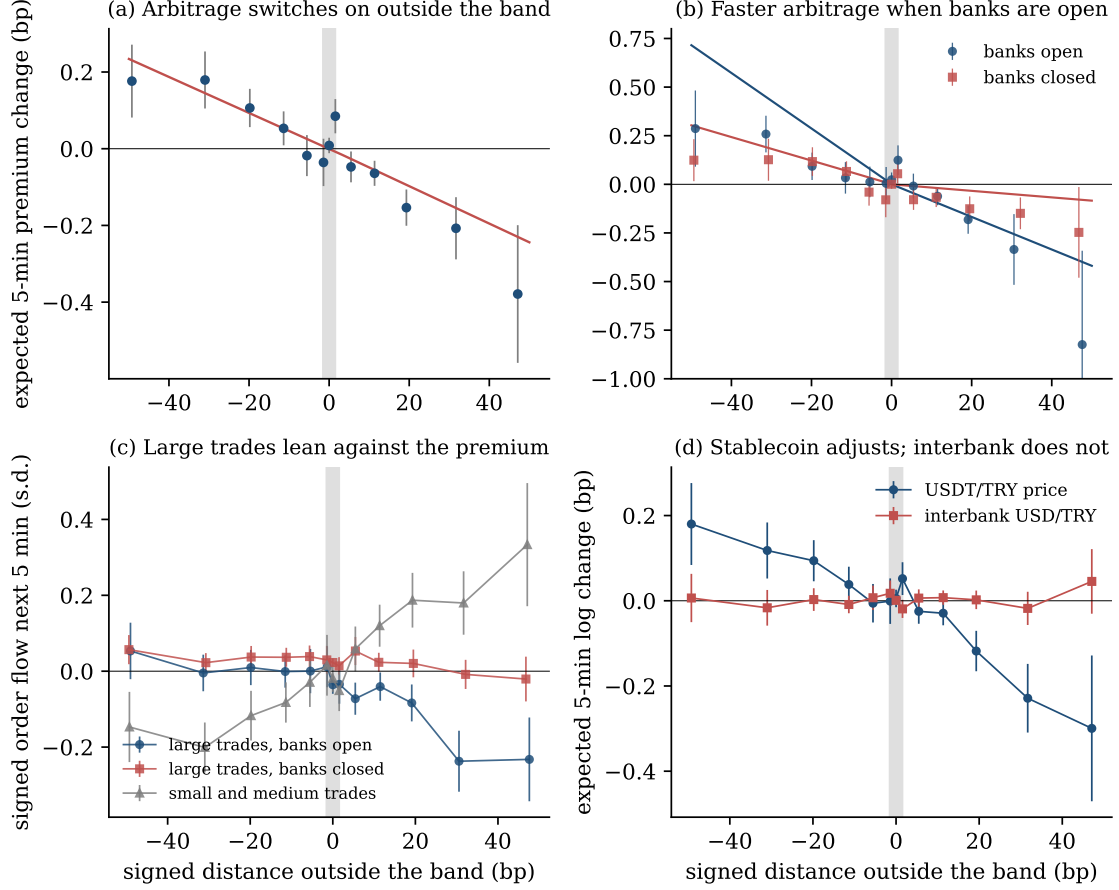


Figure 1: The mechanism in four panels. USDT/TRY premium over the interbank rate, five-minute intervals, January 2024 – August 2026. The horizontal axis is the signed distance of the premium outside its no-arbitrage band (estimated by month and banking state); all observations inside the band are plotted at zero (grey). (a) Expected change in the premium over the next five minutes, net of order flow and month effects, with the fitted threshold model. (b) The same for Istanbul business hours and for closed-bank hours. (c) Signed order flow over the next five minutes, by trade size: large trades ($\geq 10,000$ USDT) when banks are open and closed, and small and medium trades. (d) Five-minute changes in the log USDT/TRY price and the log interbank USD/TRY rate. 95% confidence intervals clustered by day.

tions here sit in the local-flat leg and its banking rails. [Baulkaran et al. \(2026\)](#) document, across nine exchanges at ten-minute frequency, that USDT liquidity, efficiency and peg corrections follow NYSE trading hours; this paper derives the band and effective capacity from the arbitrageur’s problem and identifies how banking access moves the speed of arbitrage, with bank-state-specific bands, trade-level flows and a holiday placebo. [Li et al. \(2026\)](#) study the USDC de-peg of March 2023, when banking settlement stopped over a weekend, with minute data and graph-based arbitrage detection, and find a strong link between arbitrage opportunities and banking hours; the evidence here comes from two and a half years of a

local-currency market in which banking access switches on and off every day. Relative to this literature, the paper provides a microstructure model in which a single state variable, effective arbitrage capacity, governs both the mapping from order flow to the premium and the transmission of local information to the global price, and identifies that state variable at minute frequency from prices and trades.

Stablecoin stability and arbitrage. Lyons and Viswanath-Natraj (2023) study the arbitrage that keeps stablecoins close to their peg. Ma et al. (2025) show that concentrating primary-market arbitrage in a few authorized intermediaries trades off price stability in normal times against run risk. The network result in Appendix B is the secondary-market, cross-venue analogue of their trade-off. Aldasoro et al. (2026) show that stablecoin inflows spill over into FX markets and widen parity deviations, and their model features constrained intermediaries whose capital is depleted by losses. Their evidence is a daily panel of cross-country spillovers. This paper identifies the arbitrage technology itself at minute frequency: the inaction band, the speed of adjustment, the transmission of order flow, and the banking-rail state that governs convergence.

Limits to arbitrage and slow-moving capital. The arbitrageur problem builds on Shleifer and Vishny (1997), Gromb and Vayanos (2002), Brunnermeier and Pedersen (2009), Mitchell et al. (2007) and Duffie (2010). Deviations from covered interest parity (Du et al., 2018) are a leading example in traditional markets of arbitrage-capacity-driven basis. The band structure echoes the commodity-points literature in international economics (Dumas, 1992; Obstfeld and Taylor, 1997) and threshold cointegration (Balke and Fomby, 1997; Hansen and Seo, 2002). Here the threshold is derived from an arbitrageur’s optimization and connected to order flow and price discovery.

Fragmentation and price discovery. The price-discovery result complements the information-share approach of Hasbrouck (1995) and Gonzalo and Granger (1995), which is built on linear error correction. Under threshold arbitrage, the contribution of a venue to price discovery depends on the size of the information it receives, so linear information shares average over economically distinct regimes. Lehar et al. (2024) show how fixed costs fragment liquidity across fee tiers within a decentralized exchange. This paper studies fragmentation across currencies and its implications for convergence.

The remainder of the paper is organized as follows. Section 2 presents the model and Section 3 its predictions. Section 4 describes the data and the empirical strategy, and Section 5 presents the evidence. Section 6 concludes. The appendices contain proofs, further theoretical results (the stationary law of the premium, capacity depletion and a multi-venue extension), evidence around a 2025 Turkish transfer regulation, and Monte Carlo evidence on the estimator.

2 Model

2.1 Markets and the premium

There are two markets for the same dollar stablecoin. In the *local* market the stablecoin trades against a local currency; in the *global* market it trades against the U.S. dollar (or a dollar benchmark such as a deep USDT/USD or USDC/USDT book). Let P_t^L denote the local price converted into dollars at a reference exchange rate, and P_t^G the global dollar price. The object of interest is the log premium

$$b_t \equiv \log P_t^L - \log P_t^G.$$

In a frictionless integrated market $b_t \equiv 0$. Which exchange rate is used for the conversion determines the economic interpretation. Converting at the official rate makes b_t a measure of the parallel-market premium, while converting at a contemporaneous local market rate (for example, a local BTC/USDT cross) isolates the stablecoin-specific basis. The model applies to either.

Time is continuous. Three forces move the premium.

Non-arbitrage order flow. End users (households seeking dollar exposure, importers, remitters) generate signed order flow q_t in the local market, measured as buyer-initiated minus seller-initiated volume. Absent arbitrage, order flow moves the premium with primitive impact $\beta > 0$.

Arbitrage flow. A competitive arbitrage sector chooses a flow a_t (units per unit time). A positive a_t sells the stablecoin locally and buys it globally, which lowers the premium. Arbitrage moves the local price with impact λ_L and the global price with impact λ_G , so its effect on the premium is $\lambda \equiv \lambda_L + \lambda_G$.

Noise. Everything else, including transient liquidity shocks and quote noise, is captured by σdW_t , where W is a standard Brownian motion.

2.2 The arbitrage sector

Assumption 1 (Arbitrage technology). Arbitrage in the positive (negative) direction incurs a proportional cost $c^+ > 0$ ($c^- > 0$) per unit, reflecting fees, on/off-ramp costs, FX access and transfer frictions. Holding a flow of size $|a|$ incurs a convex balance-sheet and inventory cost $a^2/(2K^\pm)$, where $K^\pm > 0$ is the *arbitrage capacity* in each direction. The arbitrage

sector chooses a_t to maximize flow profit

$$\Pi(a; b) = \begin{cases} ab - c^+ a - \frac{a^2}{2K^+}, & a \geq 0, \\ ab + c^- a - \frac{a^2}{2K^-}, & a < 0. \end{cases} \quad (1)$$

The proportional cost $c^\pm$ captures every friction that scales with the size of the trade: exchange and network fees, the spread paid on the fiat leg, and the expected cost of on- and off-ramps. Capacity $K^\pm$ captures everything that makes the *marginal* unit costlier: finite capital, inventory risk while transfers settle, counterparty limits at banks and payment providers, and withdrawal limits at exchanges. Allowing both to depend on direction is essential in capital-controlled economies. Buying stablecoins with local currency and moving the dollars abroad is generally harder than the reverse, so one expects $c^+ \neq c^-$ and $K^+ \neq K^-$.

Proposition 1 (Optimal arbitrage). *Under Assumption 1, the unique optimal arbitrage flow is*

$$a^*(b) = K^+ (b - c^+)_+ - K^- (-b - c^-)_+. \quad (2)$$

In particular, $a^(b) = 0$ for $b \in [-c^-, c^+]$; outside this no-arbitrage band, arbitrage flow is linear in the excess premium with slope equal to capacity. In the symmetric case, $a^*(b) = K \operatorname{sign}(b) (|b| - c)_+$.*

The policy is the soft-thresholding operator familiar from statistics. Its two parameters play distinct economic roles. The friction c determines *whether* arbitrage is active; capacity K determines *how strongly* it responds once active. Figure 2(a) plots the policy.

Remark 1 (Static versus dynamic arbitrage). Proposition 1 treats arbitrage as a flow decision. In a fully dynamic model with fixed costs, the option value of waiting widens the band beyond the static cost, as in Dumas (1992). The structure of the solution (inaction inside a band, active trading outside it) is preserved. The band $c^\pm$ estimated in Section 4 should therefore be read as an *effective* threshold that includes any such option value. More generally, for any convex cost $C(a)$ with $C'(0^+) = c^+$ and $C'(0^-) = -c^-$, the optimal policy is $a^*(b) = (C')^{-1}(b)$ outside $[-c^-, c^+]$ and zero inside, so all qualitative results below survive. Only the closed forms rely on the quadratic specification.

Remark 2 (Settlement latency and the location of the band). Cross-border stablecoin arbitrage is not instantaneous. Fiat legs settle with a delay, exchanges can hold withdrawals, and transfers may be subject to compliance reviews. Suppose a round trip ties up capital for a period τ , during which the arbitrageur holds dollar-denominated inventory rather than local-currency deposits. The arbitrageur then forgoes the interest differential $(i - i^*)$ over τ

in *both* directions, so the band becomes $[-c^- - (i - i^*)\tau, c^+ + (i - i^*)\tau]$. In a high-interest-rate economy this term is large. At a 45% annual lira rate, a 48-hour withdrawal hold adds roughly 25 bp to each edge of the band. Directional frictions, such as limits on outbound stablecoin transfers, raise only one edge. They therefore shift the band's location and, by Proposition 5(b), the average premium. Empirically, we estimate the edges $\ell \equiv -c^-$ and $u \equiv c^+$ freely, so the band need not contain zero.

2.3 Premium dynamics

Combining non-arbitrage order flow, optimal arbitrage and noise, the premium satisfies

$$db_t = \left(\beta q_t - \lambda \left[K^+ (b_t - c^+)_+ - K^- (-b_t - c^-)_+ \right] \right) dt + \sigma dW_t. \quad (3)$$

We write $\kappa^\pm \equiv \lambda K^\pm$ and call it *effective arbitrage capacity*: the speed at which arbitrage closes a dislocation, which combines the balance-sheet capacity $K^\pm$ of the arbitrage sector with the price impact λ of its trades. Effective capacity is the object the data identify; K and λ are not separately identified, and none of the results below requires them to be. We write

$$\mu(b) \equiv -\kappa^+ (b - c^+)_+ + \kappa^- (-b - c^-)_+$$

for the arbitrage component of the drift; Equation (3) is a threshold diffusion. Its drift is identically zero on the band, linear outside it, and kinked at $-c^-$ and c^+ . Since μ is Lipschitz, (3) has a unique strong solution for any adapted, locally integrable order-flow process q .

3 Predictions

3.1 Inaction and state-dependent reversion

Proposition 2 (State-dependent reversion). *Let $q \equiv 0$.*

- (a) *Inaction. For $b_0 \in (-c^-, c^+)$, let τ be the first exit time from the band. The stopped premium $b_{t \wedge \tau}$ is a martingale, and*

$$\mathbb{E}[\tau \mid b_0] = \frac{(c^+ - b_0)(b_0 + c^-)}{\sigma^2}.$$

- (b) *Exponential correction outside the band. If $\sigma = 0$ and $b_0 > c^+$, then $b_t = c^+ + (b_0 -$*

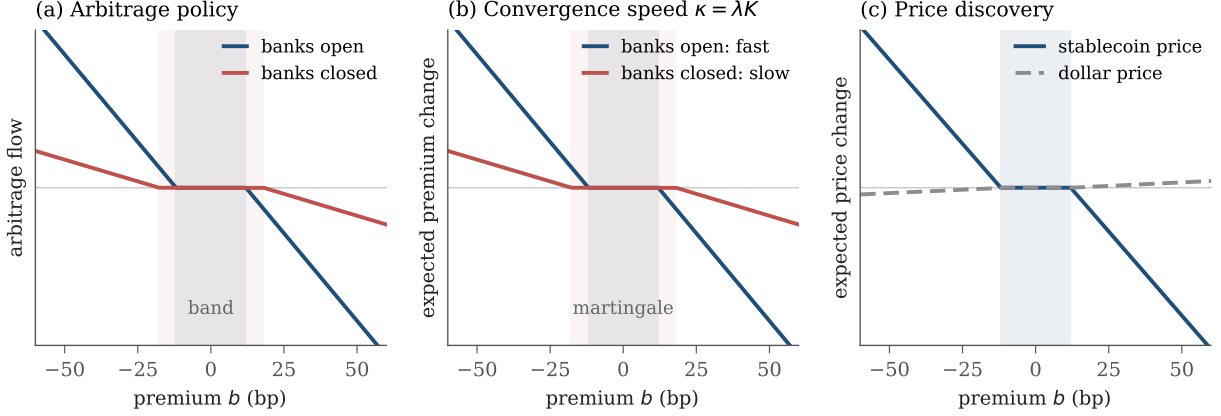


Figure 2: The model's predictions. Illustrative parameters. (a) Optimal arbitrage flow toward the band, $-a^*(b)$ (Proposition 1): zero inside the no-arbitrage band, linear outside it with slope proportional to capacity K ; closed banks lower K and may widen the band. (b) Expected change in the premium (Proposition 2): a martingale inside the band, reversion at speed $\kappa = \lambda K$ outside it. (c) The correction split between the stablecoin price and the dollar price in proportions $1 - \omega$ and ω (Proposition 4), drawn for a deep dollar market (ω small). Figure 1 shows the empirical counterparts.

$c^+)e^{-\kappa^+t}$. The excess premium has half-life

$$t_{1/2}^+ = \frac{\ln 2}{\lambda K^+},$$

and symmetrically $t_{1/2}^- = \ln 2 / (\lambda K^-)$ for $b_0 < -c^-$.

(c) Large dislocations revert faster. The proportional reversion speed $\rho(b) \equiv -\mu(b)/b$ equals zero on the band and $\kappa^+(1 - c^+/b)$ for $b > c^+$ (symmetrically for $b < -c^-$). It is non-decreasing in $|b|$ and converges to $\kappa^\pm$ as $|b| \rightarrow \infty$.

Proposition 2 contains the first set of testable predictions. Small premia do not mean-revert; they wander until they hit the edge of the band, with an expected waiting time that rises with the square of the band width. Large premia revert, and do so proportionally faster the larger they are. A linear error-correction model imposes a constant ρ . Any constant necessarily overstates reversion for small dislocations and understates it for large ones. The half-life formula gives capacity a direct, observable counterpart: halving capacity doubles the half-life of a dislocation.

3.2 Order flow, price impact and volume

Proposition 3 (Long-run price impact). Let $\sigma = 0$ and $q_t \equiv q$ constant.

(a) If $q > 0$, then from any initial condition $b_t \rightarrow b^*$, where

$$b^* = c^+ + \frac{\beta q}{\lambda K^+}.$$

If $q < 0$, $b_t \rightarrow -c^- + \beta q/(\lambda K^-)$. If $q = 0$, b_t converges to the closest point of the band.

(b) The long-run price impact of order flow is

$$\frac{\partial b^*}{\partial q} = \frac{\beta}{\lambda K^\pm},$$

which is decreasing in capacity.

(c) In the steady state, arbitrage flow equals $a^* = \beta q/\lambda$, which does not depend on $(c^\pm, K^\pm)$. Total local volume, $|q| + |a^*|$, is therefore independent of arbitrage capacity, while the premium is decreasing in it.

Part (c) is the paper's central observational-equivalence result. In steady state arbitrageurs must absorb the full non-arbitrage flow, scaled by relative impact, whatever their capacity. Capacity determines only the premium at which they are willing to do so. Volume reveals demand; the premium reveals capacity.

Corollary 1 (Volume, premia and demand across markets). *Consider two local markets $i \in \{A, B\}$ with parameters $(\beta_i, \lambda_i, c_i, K_i)$ and persistent order flow $q_i > 0$. Then*

$$b_A^* - c_A = b_B^* - c_B \iff \frac{\beta_A q_A}{\lambda_A K_A} = \frac{\beta_B q_B}{\lambda_B K_B}.$$

In particular, equal excess premia are consistent with arbitrarily different demand, provided capacity differs in proportion. Conversely, equal demand and impact ($\beta_A q_A = \beta_B q_B$, $\lambda_A = \lambda_B$) produce excess premia in inverse proportion to capacity.

Corollary 1 explains why comparisons of premia across countries, or across episodes within a country, cannot by themselves be read as comparisons of demand. Separating demand from capacity requires the joint dynamics of the premium and order flow, which is exactly what the estimator of Section 4 uses.

3.3 Price discovery: who discovers the dollar?

We now ask how information that arrives first in the local market is transmitted to the global price. Let the local price jump by $\delta > 0$ at $t = 0$ because informed local order

flow reveals news about the common dollar value (for example, news about the issuer, or about the dollar's local purchasing power that global participants learn only through prices). Before the shock $b_{0-} = 0$. Arbitrage moves the local price by $-\lambda_L a dt$ and the global price by $+\lambda_G a dt$. Define the global leg's share of arbitrage impact $\omega \equiv \lambda_G/(\lambda_L + \lambda_G)$.

Proposition 4 (Local information transmission). *Let $\sigma = 0$ and $q_t \equiv 0$ for $t > 0$. The global price at horizon h satisfies*

$$S_h(\delta) \equiv \frac{\log P_h^G - \log P_0^G}{\delta} = \omega \left(1 - \frac{c^+}{\delta}\right)_+ (1 - e^{-\lambda K^+ h}).$$

Consequently:

- (a) *local information shocks smaller than the band ($\delta \leq c^+$) are never transmitted to the global price;*
- (b) *the long-run share $S_\infty(\delta) = \omega(1 - c^+/\delta)_+$ is decreasing in the friction and independent of capacity;*
- (c) *at any finite horizon, $S_h(\delta)$ is increasing in capacity, and the half-life of transmission is $\ln 2/(\lambda K^+)$.*

Frictions determine how much local information ever reaches the global price; capacity determines how fast it gets there (Figure 9 in Appendix B). A market with heavy trading but a wide band, or one whose arbitrage capacity is impaired, can be very active and still contribute almost nothing to the price of the dollar elsewhere. This is consistent with the finding of Makarov and Schoar (2020) that more segmented, higher-premium venues contribute less to global price discovery, and it explains that finding through (c, K) . A further implication concerns measurement. Linear information-share measures (Hasbrouck, 1995; Gonzalo and Granger, 1995) average $S_h(\delta)$ over the distribution of shock sizes. They therefore depend on the volatility regime and can change sharply with no change in market structure.

4 Data and empirical strategy

Turkey is a natural laboratory for the model. Domestic demand for dollar assets is strong and persistent, the lira experienced a large political shock during the sample, policy rates were high enough to make the carry cost of settlement latency in Remark 2 economically large, and access to the fiat leg of the arbitrage runs through domestic banking rails whose availability varies over the day.

Table 1: Summary statistics: USDT/TRY, January 2024 – August 2026

	Mean	SD	P1	P50	P99	N
Premium b_t (bp), 1-min	18.43	56.43	-60.06	5.76	269.96	808,473
Premium (bp), 5-min average	18.30	56.34	-60.22	5.62	269.74	162,624
USDT/TRY volume (thousand USDT per min)	64.91	129.97	1.13	36.42	543.61	808,473
Signed order flow (thousand USDT per min)	9.12	112.69	-212.43	2.46	210.12	808,473
Trades per minute	143.40	204.96	8.00	90.00	952.00	808,473
Interbank USD/TRY bid-ask spread (bp)	1.26	1.09	0.50	0.97	6.74	15,149
Triangular basis USDT/TRY vs. BTC cross (bp)	-0.13	2.38	-6.18	-0.10	5.68	1,378,099

Notes: One-minute data on weekdays, excluding 20:30–22:30 UTC. The premium is the log difference between the Binance USDT/TRY minute VWAP and the Dukascopy interbank USD/TRY mid. Signed order flow is taker-buy minus taker-sell volume. The triangular basis compares USDT/TRY with the cross rate implied by BTC/TRY and BTC/USDT on the same exchange (all days).

Local market. One-minute USDT/TRY candles come from Binance’s public data archive, from 2 January 2024 to 31 August 2026. For each minute the archive reports volume, the number of trades, quote volume and *taker-buy* volume. The local price P_t^L is the minute volume-weighted average price (quote volume divided by volume). Signed order flow is taker-buy minus taker-sell volume, $OFI_t = 2 TBV_t - V_t$, in USDT. The market is deep: the median minute has 90 trades and 36 thousand USDT of volume (Table 1). Binance aggregated-trade files record every aggressive trade with its size and side, which is used in Section 5 to split order flow by trade size.

Benchmark. The dollar price is the USD/TRY mid-quote from one-minute candles of Dukascopy’s SWFX marketplace, an institutional FX venue that aggregates quotes from bank and non-bank liquidity providers; its median relative bid-ask spread is about 1 bp. For brevity it is called the interbank rate below. The sample uses minutes with interbank activity on weekdays, excludes the 20:30–22:30 UTC rollover window, and discards minutes with $|b_t| > 10\%$. The premium is $b_t = 10^4(\log P_t^L - \log P_t^G)$ in basis points.

Where the friction sits. On the same exchange, the USDT/TRY price and the USDT price implied by the BTC/TRY and BTC/USDT order books differ with a standard deviation of 2.4 bp; this triangular arbitrage moves no fiat money. The premium over the interbank rate has a standard deviation of 56 bp. The relevant frictions are those of the fiat and cross-border leg, which is what c and K measure.

Estimating equation. The premium is averaged over non-overlapping five-minute intervals, which removes most of the bid-ask bounce in the local price, and order flow is summed over the same interval and standardized by its within-month standard deviation. An Euler

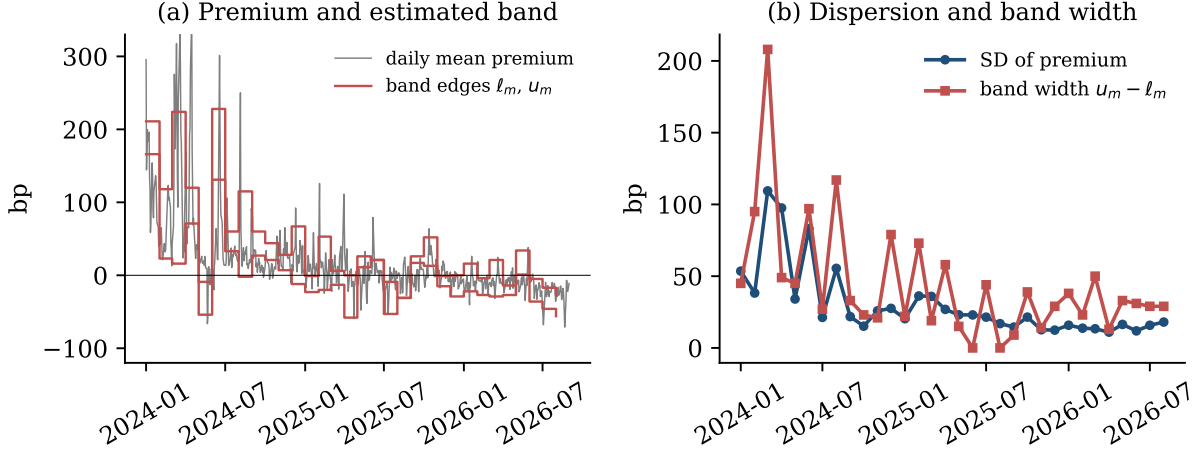


Figure 3: The USDT/TRY premium over the interbank rate. Panel (a): daily mean premium and the estimated monthly band edges ℓ_m and u_m . Panel (b): monthly standard deviation of the five-minute premium and the width of the estimated band.

discretization of (3) gives

$$\Delta b_t = \alpha_m + \beta OFI_t - \kappa^+ (b_{t-1} - u_m)_+ + \kappa^- (\ell_m - b_{t-1})_+ + \varepsilon_t, \quad (4)$$

where t indexes five-minute intervals, α_m is a month fixed effect, $[\ell_m, u_m]$ is the month's no-arbitrage band, and $\kappa^\pm = \lambda K^\pm \Delta$ is effective capacity per interval. The structural objects are the band, the speeds $\kappa^\pm$, the primitive impact β (in bp per one-standard-deviation five-minute order-flow shock) and the long-run impact $\beta/\kappa^\pm$ of Proposition 3. Because K and λ enter only through their product, the data identify effective arbitrage capacity κ , not balance-sheet capacity K separately.

Month-specific bands. The level of the premium moves at low frequency (Figure 3): it averaged more than 100 bp in the first half of 2024, fluctuated around zero in 2025, and turned into a discount of 10–25 bp in 2026. Remark 2 attributes such movements to carry and directional frictions, which shift the band. The edges (ℓ_m, u_m) are therefore profiled month by month over a 1 bp grid, subject to $\ell_m \leq u_m$ and 5% trimming. Given the bands, the slopes of (4) are estimated by pooled OLS. Thresholds are super-consistent (Hansen, 2000), so slope inference treats them as known. Standard errors are clustered by day. Linearity is tested against the threshold model with the sup-LR statistic (Hansen, 1996; Hansen and Seo, 2002), summed across months, with a day-clustered wild bootstrap under the linear null. Appendix D shows that the profiled estimator recovers the band, the speeds and the order-flow impact at sample sizes of two weeks of minute data.

5 Evidence

5.1 Threshold dynamics

Table 2 reports the main estimates. Column (1) is a linear error-correction model with month fixed effects; it attributes slow, uniform reversion to the premium ($\hat{\kappa} = 0.0039$ per five minutes). Column (2) estimates the threshold model. Outside the band the premium reverts at $\hat{\kappa}^+ = 0.0054$ above it and $\hat{\kappa}^- = 0.0085$ below it, both significant at 1%. Column (3) adds a linear reversion term that is active only inside the band; its coefficient is -0.0006 with a standard error of 0.0007 . Inside the band the premium is a martingale, as Proposition 2(a) predicts. The median band is 32 bp wide and contains 39% of the intervals. Summed across months, the sup-LR statistic is 464, and the wild bootstrap rejects linear error correction with a p -value of 0.01, the smallest attainable with 99 draws. Figure 1(a) shows the threshold shape non-parametrically, and Table 3 reports reversion by size of dislocation: zero inside the band, and 0.5–0.6% per five minutes beyond 10 bp outside it. Very small estimated excursions outside the band are noisy around the profiled threshold; reversion becomes monotone and economically meaningful once the dislocation exceeds roughly 10 bp.

5.2 Banking rails and effective capacity

During Istanbul business hours, banks process transfers and the fiat leg of the arbitrage can be executed; outside them, the stablecoin market keeps trading while the fiat leg slows down. Column (4) of Table 2 interacts the reversion terms and order flow with an indicator for 09:00–17:00 Istanbul time. Above the band, the speed of arbitrage is 0.0025 outside business hours and 0.0104 during them; the difference, 0.0079 (s.e. 0.0014), is a fourfold increase. The primitive impact of order flow is statistically the same in the two states (0.64 and 0.58). By Proposition 3, the long-run impact of a sustained one-standard-deviation order-flow imbalance is $\beta/\kappa^+ \approx 56$ bp during business hours and 257 bp outside them. This is Corollary 1 inside a single market: the same demand moves the premium nearly five times as much when effective capacity is low.

Friction or capacity? Closed banks could also widen the band, since settlement latency enters the band through carry (Remark 2). Column (1) of Table 4 re-estimates the band separately for open and closed hours within each month, with month-by-state fixed effects. Closed-hour bands are wider on average, 48 bp against 40 bp (medians 34 and 24 bp), consistent with longer settlement. With the friction allowed to differ, effective capacity above the band is about five times higher when banks are open: the increment is 0.0067 (s.e.

Table 2: Threshold arbitrage in the USDT/TRY premium

	(1)	(2)	(3)	(4)
OFI_t	0.603*** (0.030)	0.604*** (0.030)	0.604*** (0.030)	0.635*** (0.032)
$-(b_{t-1} - u_m)_+ [\kappa^+]$		0.0054*** (0.0013)	0.0054*** (0.0013)	0.0025* (0.0013)
$(\ell_m - b_{t-1})_+ [\kappa^-]$		0.0085*** (0.0025)	0.0085*** (0.0025)	0.0074*** (0.0018)
$-b_{t-1} [\text{linear } \kappa]$	0.0039*** (0.0006)			
$-(b_{t-1} - m_m) \cdot \mathbb{I}\{\text{in band}\}$			-0.0006 (0.0007)	
$OFI_t \times BH$				-0.052 (0.033)
$-(b_{t-1} - u_m)_+ \times BH$				0.0079*** (0.0014)
$(\ell_m - b_{t-1})_+ \times BH$				0.0023 (0.0018)
Month fixed effects	Yes	Yes	Yes	Yes
Month-specific band $[\ell_m, u_m]$	No	Yes	Yes	Yes
N	162,624	162,624	162,624	162,624
Within R^2	0.042	0.042	0.042	0.043

Notes: Dependent variable: five-minute change in the premium (bp). OFI is standardized within month. $[\ell_m, u_m]$ is the band estimated month by month; m_m is its midpoint. BH is an indicator for Istanbul business hours (09:00–17:00). Standard errors clustered by day in parentheses. ***, **, * denote significance at 1%, 5%, 10%.

Table 3: Reversion by size of dislocation

	Mean distance (bp)	Reversion per 5 min, $\hat{\rho}$	Half-life (min)	N
Inside band, inner half	6.9	-0.0028 (0.0020)	∞	32,221
Inside band, outer half	21.6	-0.0020 (0.0006)	∞	31,020
Outside, excess $\in (0, 5]$ bp	2.5	-0.0154 (0.0055)	∞	20,099
Outside, excess $\in (5, 10]$ bp	7.4	0.0004 (0.0023)	7868	18,562
Outside, excess $\in (10, 20]$ bp	14.5	0.0052 (0.0011)	662	24,682
Outside, excess $\in (20, 40]$ bp	28.4	0.0058 (0.0008)	593	19,918
Outside, excess > 40 bp	81.9	0.0037 (0.0006)	949	16,122

Notes: Proportional reversion $\hat{\rho}$ is the sum of reversion toward the band (net of $\hat{\beta} OFI_t$) divided by the sum of distances, measured from the band midpoint inside the band and from the nearest edge outside it. Delta-method standard errors clustered by day. Half-lives in minutes.

0.0023) on a closed-hours speed of 0.0017. Banking hours raise effective arbitrage capacity, and the result is not explained by a wider no-arbitrage band. Column (2) estimates β from in-band intervals only, where the model implies $a^* = 0$, which allows β to be estimated without the model-implied arbitrage component; it is 0.64 in both states, and holding it fixed leaves the speeds unchanged.

Table 4: Separating the friction from effective capacity

	State-specific bands (1)	In-band β , fixed (2)
κ^+ , banks closed	0.0017 (0.0015)	0.0017 (0.0015)
$\times$ banks open	0.0067*** (0.0023)	0.0067*** (0.0023)
κ^- , banks closed	0.0061*** (0.0012)	0.0061*** (0.0012)
$\times$ banks open	0.0082 (0.0054)	0.0082 (0.0054)
β , banks closed	0.632*** (0.032)	0.641*** (0.039)
$\times$ banks open	-0.047 (0.035)	-0.004 (0.053)
Mean band width, banks open (bp)	39.6	
Mean band width, banks closed (bp)	47.6	
Difference, closed – open (bp)	8.0 (8.7)	
N	162,624	162,624

Notes: Column (1): bands $[\ell, u]$ profiled separately for each month and banking state (open: 09:00–17:00 Istanbul), month-by-state fixed effects, all slopes interacted with the banks-open indicator. Column (2): β estimated on in-band intervals only and held fixed; the dependent variable is $\Delta b_t - \hat{\beta} OFI_t$. Band widths are averages of the monthly estimates; the standard error of the difference is across months. Standard errors clustered by day.

The interbank benchmark. Table 5 keeps only intervals in which the interbank market is actively quoted (column 2); the business-hours increase in κ^+ is 0.0088 (s.e. 0.0016). Columns (3) and (4) split the correction into its two legs. Noisy night-time interbank quotes would show up as night-time reversion in the interbank leg. The interbank leg shows no reversion in either state, and the business-hours effect is detectable only in the stablecoin price.

The banking day and a holiday placebo. Figure 4(a) estimates the speed of arbitrage by two-hour block of the Istanbul day. It is indistinguishable from zero from midnight until 09:00, jumps after banks open, and stays positive through the evening. Time of day alone could produce such a pattern through liquidity or attention. Weekday public holidays separate the clock from the banks: the stablecoin market trades and the clock reads business hours, but banks are closed. On the 29 weekday holidays in the sample, the speed of arbitrage between 09:00 and 17:00 is 0.0029 (s.e. 0.0018), at its night-time level and far below the 0.018–0.025 estimated for the blocks from 10:00 to 16:00 on ordinary weekdays. In the full specification, the business-hours increment is 0.0071 (s.e. 0.0021) on ordinary weekdays and -0.0000 (s.e. 0.0012) on holidays.

Table 5: Banking hours and the interbank benchmark

	Δb_t		$\Delta \log P_t^L$	$\Delta \log P_t^G$
	All (1)	Active interbank (2)	All (3)	All (4)
$-(b_{t-1} - u_m)_+ [\kappa^+, \text{banks closed}]$	0.0025* (0.0013)	0.0016 (0.0017)	0.0024* (0.0013)	-0.0000 (0.0004)
$\times \text{banks open}$	0.0079*** (0.0014)	0.0088*** (0.0016)	0.0074*** (0.0016)	-0.0006 (0.0004)
$(\ell_m - b_{t-1})_+ [\kappa^-, \text{banks closed}]$	0.0074*** (0.0018)	0.0081*** (0.0023)	0.0059*** (0.0011)	-0.0015 (0.0015)
$\times \text{banks open}$	0.0023 (0.0018)	0.0028 (0.0021)	0.0008 (0.0007)	-0.0016 (0.0016)
$\text{OFI}_t [\text{banks closed}]$	0.635*** (0.032)	0.699*** (0.038)	0.645*** (0.032)	0.007 (0.008)
$\times \text{banks open}$	-0.052 (0.033)	-0.058 (0.040)	-0.061* (0.034)	-0.006 (0.025)
N	162,624	111,037	162,624	162,624

Notes: Specification (4) with all coefficients interacted with an indicator for Istanbul business hours. Column (2) keeps intervals whose interbank tick activity exceeds the 25th percentile of business-hours intervals. Columns (3) and (4) use the five-minute change in the log USDT/TRY price and in the log interbank rate (bp) as dependent variables. Month fixed effects and month-specific bands throughout; standard errors clustered by day.

5.3 A cross-country contrast: rails that never close

If closed banking rails drive the night-time collapse of effective capacity, the collapse should be specific to markets whose large-value rails close. In Turkey, large transfers between banks run through EFT, which operates during business hours on business days; the instant FAST system runs around the clock but caps each transfer (at 100,000 lira for individuals and 250,000 lira for commercial accounts in 2025–2026, a few thousand dollars). In Mexico, SPEI, the real-time gross settlement system of the central bank, runs 24 hours a day on every day of the year and settles high-value transfers (Banco de México, 2026). Binance lists USDT against the Mexican peso and Dukascopy quotes USD/MXN, so the identical construction applies to both markets over the common window November 2024 – August 2026, with business hours 09:00–17:00 local time in each.

Table 6 and Figure 5 report the comparison. Levels of the speed are not comparable across markets, because the price impact of arbitrage flow λ differs with depth, and the Mexican market trades about one fiftieth of the Turkish volume. The open-to-closed ratio removes λ under state-invariance within each market. It is 4.2 in Turkey and 1.6 in Mexico, and the difference in log ratios, 0.94 (s.e. 0.47), is significant at 5%. At night, arbitrage in Mexico retains nearly two thirds of its daytime speed; in Turkey it retains about one twentieth. The bands tell the same story: closed-hours bands are 8 bp wider than open-

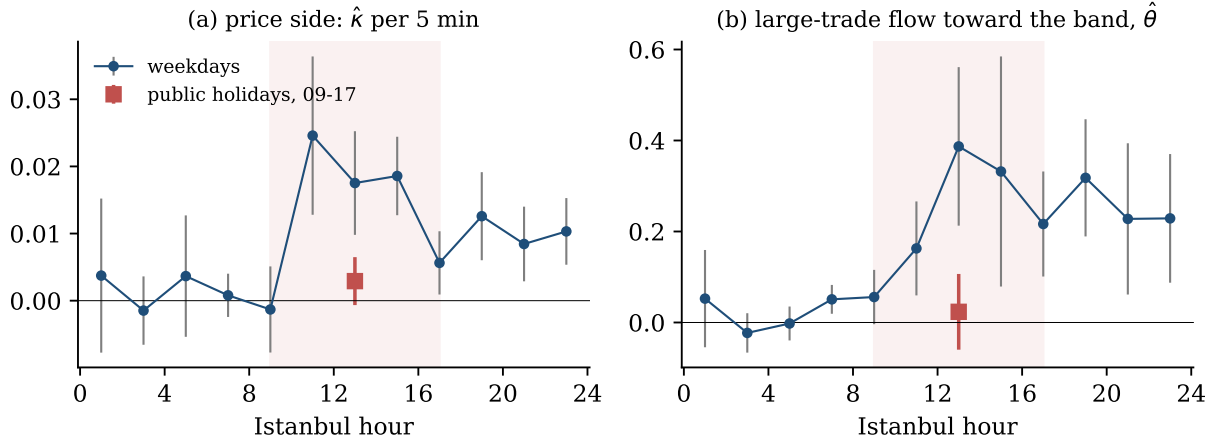


Figure 4: The banking day and a holiday placebo. (a) Speed of arbitrage outside the band, $\hat{\kappa}$, by two-hour block of the Istanbul day on ordinary weekdays, and during 09:00–17:00 on the 29 weekday public holidays in the sample (square). (b) The same for the large-trade response $\hat{\theta}$ of Table 8, column (2), in standard deviations of flow per 100 bp. Bands are estimated by month and banking state; month fixed effects; 95% confidence intervals clustered by day. The shaded area marks business hours.

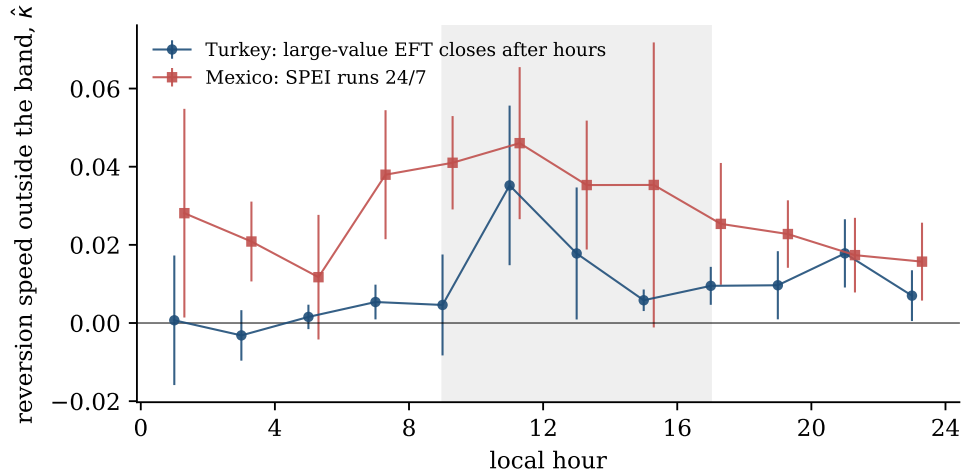


Figure 5: The daily cycle of arbitrage in Turkey and Mexico. Speed of arbitrage outside the band, $\hat{\kappa}$, by two-hour block of local time, November 2024 – August 2026, with month fixed effects and 95% confidence intervals clustered by day. The shaded area marks local business hours.

hours bands in Turkey and 2 bp wider in Mexico. The Mexican daytime increment shows that part of the intraday variation is common to both markets, as daytime FX liquidity and staffing would produce; the holiday placebo ties the Turkish increment to banks being open. The night-time collapse of effective capacity is specific to the market whose large-value rails close.

Table 6: Turkey and Mexico: rails that close and rails that do not

	Turkey (EFT closes after hours)	Mexico (SPEI 24/7)	Difference
Speed outside band, banks closed	0.0052** (0.0021)	0.0241*** (0.0025)	
× business hours	0.0164** (0.0071)	0.0149*** (0.0058)	0.0015 (0.0092)
In-band placebo, banks closed	−0.0052*** (0.0016)	−0.0046 (0.0041)	
× business hours	0.0038 (0.0054)	−0.0007 (0.0077)	
Above band only: $\kappa^+ \times$ business hours	0.0086* (0.0046)	0.0131* (0.0069)	−0.0045 (0.0083)
Open/closed ratio of the speed outside the band log ratio	4.2 1.43 (0.44)	1.6 0.48 (0.17)	0.94** (0.47)
SD of premium (bp)	24.9	27.2	
Mean band width, open / closed hours (bp)	25 / 33	20 / 22	
USDT volume per minute (thousand)	51.1	0.9	
N	113,023	113,383	

Notes: USDT/TRY and USDT/MXN on Binance against Dukascopy USD/TRY and USD/MXN, November 2024 – August 2026, identical construction: five-minute averages, bands by month and local banking state, month-by-state effects. The dependent variable is $-s_t(\Delta b_t - \hat{\beta} OFI_t)$ as in Table 8, column (1); business hours are 09:00–17:00 local time (Istanbul UTC+3, Mexico City UTC−6). Log-ratio standard errors by the delta method; the cross-market difference treats the two samples as independent. Standard errors clustered by day.

5.4 Arbitrage in the trades

The model has two flows: end-user demand q , which pushes the premium, and arbitrage a^* , which leans against it once the premium leaves the band. Trade-level data reveal two sharply different flow patterns corresponding to the two forces in the model. Signed order flow is split by the size of each aggressive trade into small ($<1,000$ USDT), medium (1,000–10,000) and large ($\geq 10,000$) trades, which account for 22%, 37% and 41% of volume. Trade size does not identify traders, but it separates flows with different behavior.

Table 7 regresses each class on the lagged excess premium. Small and medium trades exhibit demand-like flow: they buy when the premium is above the band and sell when it is below it. Large trades exhibit arbitrage-like flow: they sell when the premium is above the band and buy when it is below it, and above the band their selling is about seven times stronger when banks are open. Large trades also carry the largest price impact.

Table 8 turns this into a direct test of the arbitrage policy $a^*(b) = K(|b| - c)_+$. The dependent variable is signed flow over the next five or thirty minutes, oriented toward the band, $A_t = -s_t OFI_t$, where s_t is the side of the band on which the premium stands. The

Table 7: Order flow by trade size and the excess premium

	Standardized signed order flow, by trade size		
	Small < 1k USDT	Medium 1k–10k	Large ≥ 10k
$(b_{t-1} - u_m)_+$	0.491*** (0.167)	0.141 (0.102)	−0.042* (0.022)
× banks open	0.499*** (0.186)	0.705*** (0.176)	−0.260*** (0.036)
$(\ell_m - b_{t-1})_+$	−0.800*** (0.100)	−0.693*** (0.094)	0.151*** (0.025)
× banks open	0.256*** (0.060)	0.408*** (0.071)	−0.086*** (0.022)
Share of volume	0.22	0.37	0.41
Price impact on Δb_t (bp per s.d.)	−0.020 (0.061)	0.225*** (0.019)	0.554*** (0.032)
N		162,624	

Notes: Binance USDT/TRY aggregated trades, January 2024 – August 2026. Signed volume of aggressive trades by size class, summed over five-minute intervals and standardized within month, regressed on the lagged excess premium above and below the month-specific band and its interaction with the banks-open indicator. Coefficients are in standard deviations of flow per 100 bp of excess. The price-impact row regresses Δb_t on the three standardized flows and the excess terms. Month fixed effects; standard errors clustered by day.

regressors are the distance outside the band, d_t^{out} , and, as a placebo, the distance from the band’s midpoint for premia inside it, d_t^{in} . Large-trade flow toward the band rises with d_t^{out} , by 0.030 standard deviations per 100 bp when banks are closed and by a further 0.108 when they are open; over thirty minutes the response is 0.13 and 0.83. Inside the band, large-trade flow does not respond to the premium in either state. Small and medium flow moves away from the band, inside and outside it alike.

Price and flow move together. The theory links the two sides. With $\kappa_s = \lambda K_s$ on the price side and a large-trade response $\theta_s = \eta K_s$ on the flow side, under state-invariant mappings λ and η from arbitrage activity into prices and into observed large-trade flow, a common capacity shift implies equal open-to-closed ratios on the two sides, $\kappa_{open}/\kappa_{closed} = \theta_{open}/\theta_{closed}$. Column (1) of Table 8 estimates the price-side speed on the same sample and orientation. The open-to-closed ratio is 3.3 for prices and 4.6 for large-trade flow. Estimated jointly with day-clustered cross-equation covariance, the difference in log ratios is −0.34 (s.e. 0.51), so equality is not rejected. The estimates are consistent with the restriction implied by a common shift in capacity: the price-side estimate κ and the flow-side arbitrage response move together across banking states. On weekday holidays the large-trade response stays at its closed-bank level as well (Figure 4(b)).

Table 8: The arbitrage policy in prices and in trades

	Price side	Large trades		Small and medium trades	
	$-s_t(\Delta b - \hat{\beta} OFI)$ (1)	next 5 min (2)	next 30 min (3)	next 5 min (4)	next 30 min (5)
Distance outside band, d^{out}	0.0031*** (0.0008)	0.030** (0.014)	0.128 (0.089)	-0.336*** (0.114)	-1.848*** (0.712)
× banks open	0.0071*** (0.0021)	0.108*** (0.025)	0.702*** (0.157)	-0.149 (0.222)	-0.987 (1.320)
Distance inside band, d^{in} (placebo)	0.0012 (0.0014)	-0.027 (0.035)	-0.164 (0.205)	-0.327** (0.144)	-1.964** (0.894)
× banks open	-0.0008 (0.0017)	0.032 (0.061)	0.246 (0.350)	0.252 (0.244)	1.818 (1.489)
Open/closed ratio of the d^{out} response	3.3	4.6			
Test of equal ratios, log difference (s.e.)	-0.34 (0.51)				
N	162,624	162,624	153,847	162,624	153,847

Notes: s_t is the side of the band on which b_{t-1} lies (+1 above, -1 below; for in-band premia, the side of the midpoint). Column (1): $-s_t(\Delta b - \hat{\beta} OFI_t)$. Columns (2)–(5): signed order flow of large ($\geq 10,000$ USDT) or small and medium trades over the next five or thirty minutes, oriented toward the band ($-s_t OFI$), standardized within month; coefficients per 100 bp. d^{out} is the distance outside the band and d^{in} the distance from the midpoint for in-band premia. Bands by month and banking state; “banks open” excludes weekday public holidays. The ratio test uses the joint day-clustered covariance of columns (1) and (2). Month fixed effects; standard errors clustered by day.

5.5 Who discovers the dollar?

Proposition 4 links price discovery to ω , the share of an arbitrage correction borne by the global price. Table 9 estimates (4) separately for the local price and the interbank rate. When the premium is above the band, the local price falls at rate 0.0051 and the interbank rate stays put (-0.0002 , s.e. 0.0004), so $\hat{\omega} = 0.04$; below the band the interbank coefficient is again insignificant. Stablecoin order flow moves the local price (0.61) and leaves the interbank rate unchanged (0.003, s.e. 0.018). Figure 1(d) shows the same asymmetry by distance from the band. The dollar’s lira price is discovered in the interbank market, and the stablecoin market follows it with a lag set by effective capacity.

The 19 March 2025 lira shock shows the mechanism at minute frequency (Figure 6). After the detention of Istanbul’s mayor that morning, the interbank rate jumped by up to 11.0% while USDT/TRY rose by at most 6.0%, and the premium fell to -660 bp at 07:46 UTC, during Istanbul business hours. The dislocation closed from both sides as the interbank rate retraced amid heavy official intervention and the stablecoin price rose; the premium was back inside the band within 75 minutes, and by the afternoon both prices stood 3–3.5% above their pre-shock levels.

Table 9: Which price adjusts?

	$\Delta \log P_t^L$ (USDT/TRY)	$\Delta \log P_t^G$ (USD/TRY)
OFI_t	0.609*** (0.030)	0.003 (0.018)
$-(b_{t-1} - u_m)_+$	0.0051*** (0.0012)	-0.0002 (0.0004)
$(\ell_m - b_{t-1})_+$	0.0062*** (0.0009)	-0.0024 (0.0023)
Month fixed effects	Yes	Yes
Implied global share ω (premium side)		0.04
Implied global share ω (discount side)		0.28
N	162,624	162,624

Notes: Specification (4) with the five-minute change in the log USDT/TRY price (column 1) or the log interbank USD/TRY rate (column 2), in bp, as the dependent variable. ω is the global price’s share of the total correction, $\kappa_G/(\kappa_L + \kappa_G)$. Standard errors clustered by day.

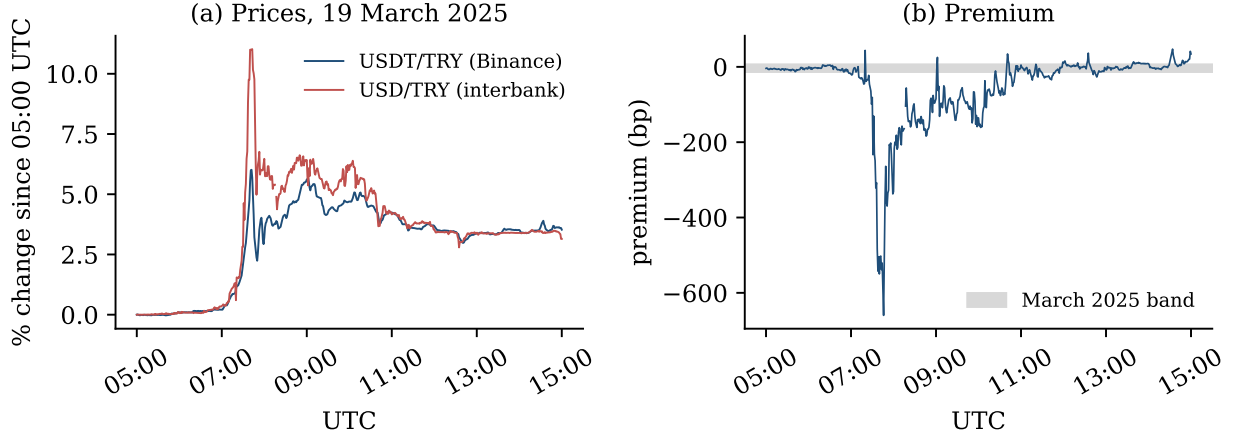


Figure 6: The 19 March 2025 lira shock. Panel (a): percentage change in USDT/TRY (Binance VWAP) and interbank USD/TRY since 05:00 UTC. Panel (b): one-minute premium and the band estimated for March 2025.

5.6 What round-the-clock banking would do

The estimates allow a counterfactual that replays history. Column (1) of Table 4 is fitted with month-by-state effects, and the realized residual of every five-minute interval is recovered. Feeding the realized residuals, order flow, bands, effects and banking states back through the model reproduces the observed path of the premium exactly. The counterfactual changes one thing: the closed-hours speeds are replaced by their open-hours values, so that effective capacity runs on the clock of the stablecoin market, and every realized shock is held fixed (Table 10). The share of time the premium spends more than 50 bp outside its band falls by 19% (90% interval from parameter uncertainty: 3–27%), the mean distance outside the band

Table 10: Counterfactual: banks open around the clock

	Observed history (exact replay)	Counterfactual: banks always open	Change [90% interval]
SD of premium around band midpoint (bp)	47.0	43.7	-7% [-9, -3]
Share of time > 25 bp outside band	0.185	0.156	-16% [-20, -6]
Share of time > 50 bp outside band	0.075	0.061	-19% [-27, -3]
Mean distance outside band (bp)	15.5	13.5	-13% [-17, -4]
Mean duration of an excursion (min)	157	151	-4% [-6, -3]
Volume-weighted wedge on small/medium trades (bp)	19.0	16.7	-12% [-15, -4]

Notes: Historical-shock replay of the model of Table 4, column (1): bands by month and banking state, month-by-state effects, state-specific order-flow impact and speeds, and the realized residual of every five-minute interval. With the estimated speeds the replay reproduces the observed path exactly (first column). The counterfactual sets the closed-hours speeds equal to the open-hours speeds and holds all realized shocks fixed. The premium is reset to its observed value at the start of each contiguous segment. Brackets: 5th and 95th percentiles of the change over 200 draws of the speeds from their day-clustered sampling distribution, with the replay re-fitted under each draw. The wedge is the distance outside the band weighted by small- and medium-trade volume.

by 13%, and the volume-weighted wedge paid on small and medium trades by 12%, or 2.3 bp. Applied to the 31 billion USDT of small and medium trades in the sample on Binance alone, this is about USD 7 million. These numbers isolate the contribution of the banking clock to the cost of dollar access in the observed history. This is a partial-equilibrium historical replay, not a general-equilibrium welfare calculation.

5.7 Integration over time and robustness

Table 11 estimates (4) by year. The primitive impact of standardized order flow falls from 1.04 bp in 2024 to 0.43 in 2025 and 0.25 in 2026, the speed of arbitrage above the band rises from 0.0051 to 0.0095, and the standard deviation of the premium falls from 74 to 16 bp: through Proposition 5, whose tail precision is $\lambda K/\sigma^2$, the tightening of the premium distribution reflects growing effective capacity. Table 12 re-estimates the main specification at one- and fifteen-minute frequencies, with last-trade prices in place of VWAPs, without the 19 March – 30 April 2025 window, with quarterly bands, and with 10% trimming. Reversion outside the band is positive and significant in every specification and scales with the sampling interval, and the business-hours increase in κ^+ is significant at 1% throughout. Inside the band, reversion is zero at the one- and five-minute frequencies with monthly bands; at the fifteen-minute frequency and with quarterly bands, which place some out-of-band observations inside the band, it is between one fifth and two fifths of the speed outside it.

Table 11: Estimates by year

Year	β	κ^+	κ^-	Median band width (bp)	SD of premium (bp)
2024	1.041*** (0.055)	0.0051*** (0.0014)	0.0050*** (0.0011)	47	74.1
2025	0.431*** (0.040)	0.0065*** (0.0015)	0.0212** (0.0091)	20	24.5
2026	0.249*** (0.018)	0.0095*** (0.0015)	0.0094*** (0.0020)	30	16.4

Notes: Specification (4) with year-specific slopes, month fixed effects and month-specific bands. Standard errors clustered by day.

Table 12: Robustness

	β	κ^+	κ^-	In-band reversion	$\kappa_{BH}^+ - \kappa_{off}^+$
Baseline: 5-min, VWAP, monthly bands, 5% trimming	0.604*** (0.030)	0.0054*** (0.0013)	0.0085*** (0.0025)	-0.0006 (0.0007)	0.0079*** (0.0014)
1-minute frequency	0.342*** (0.013)	0.0011*** (0.0003)	0.0036*** (0.0009)	-0.0005* (0.0003)	0.0014*** (0.0003)
15-minute frequency	1.116*** (0.067)	0.0187*** (0.0028)	0.0243*** (0.0070)	0.0039** (0.0018)	0.0278*** (0.0045)
Last-trade price instead of VWAP	0.643*** (0.032)	0.0056*** (0.0013)	0.0091*** (0.0025)	-0.0006 (0.0007)	0.0080*** (0.0015)
Excluding 19 Mar–30 Apr 2025	0.598*** (0.029)	0.0054*** (0.0013)	0.0062*** (0.0010)	-0.0005 (0.0007)	0.0080*** (0.0015)
Quarterly bands	0.604*** (0.030)	0.0044*** (0.0011)	0.0130*** (0.0046)	0.0017*** (0.0006)	0.0062*** (0.0015)
10% trimming	0.604*** (0.030)	0.0042*** (0.0011)	0.0098*** (0.0034)	-0.0002 (0.0007)	0.0070*** (0.0014)

Notes: Each row re-estimates columns (2)–(4) of Table 2. “In-band reversion” is the coefficient on $-(b_{t-1} - m_m)\mathbb{I}\{\text{in band}\}$ in column (3). The last column is the business-hours interaction on κ^+ in column (4). Reversion coefficients are per sampling interval. Standard errors clustered by day.

6 Conclusion

The same order-flow shock moves a local stablecoin price by different amounts at different times because effective arbitrage capacity, the speed at which arbitrage closes a dislocation, varies, and access to the banking system governs it. A per-unit cost and a finite capacity turn the local premium into a threshold diffusion, separate what volume reveals (demand) from what the premium reveals (capacity), and filter which local information reaches the global price.

The evidence triangulates the mechanism from prices, trades, cross-country variation, and price discovery. Prices: premia inside a band of about 30 bp do not revert, and above it they revert three to five times faster when banks are open, also with bands estimated by banking state. Trades: small and medium trades open dislocations, large trades lean against them in proportion to the distance outside the band, and their response rises with bank opening by a factor statistically indistinguishable from the price-side speed and stays

flat on bank holidays. Across countries: in Mexico, whose central-bank settlement system never closes, arbitrage keeps most of its speed at night. Price discovery: corrections are detectable in the stablecoin price and not in the interbank rate. Replaying history with daytime capacity around the clock shrinks the premium's large excursions by about a fifth.

The same design can be extended to the naira, won, and other local-currency stablecoin markets, where different capital-control regimes and interest differentials imply different bands for the same demand, and to direct measures of capacity such as exchange withdrawal flows. Whether a local market helps discover the dollar's value, or pays for access to it, is a question its premium and its trades can answer.

Data and code availability. All data are public: Binance spot candles and aggregated trades (<https://data.binance.vision>) and Dukascopy interbank candles. Code that downloads the data and reproduces every table and figure is available from the authors upon request.

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A Proofs

Proof of Proposition 1. $\Pi(\cdot; b)$ is continuous and strictly concave on $\mathbb{R}$: it is strictly concave on each half-line, and at $a = 0$ its left derivative $b + c^-$ exceeds its right derivative $b - c^+$, so the kink is concave. Hence the maximizer is unique and is characterized by $0 \in \partial\Pi(a^*; b)$. At $a = 0$ the superdifferential is $[b - c^+, b + c^-]$, which contains 0 if and only if $-c^- \leq b \leq c^+$. If $b > c^+$, the right derivative at zero is positive, so $a^* > 0$ solves $b - c^+ - a/K^+ = 0$, i.e. $a^* = K^+(b - c^+)$. If $b < -c^-$, the left derivative at zero is negative, so $a^* < 0$ solves $b + c^- - a/K^- = 0$, i.e. $a^* = K^-(b + c^-) = -K^-(-b - c^-)$. Combining the cases gives (2). $\square$

Proof of Proposition 2. (a) With $q \equiv 0$ and $b_s \in (-c^-, c^+)$ for $s < \tau$, the drift vanishes, so $b_{t \wedge \tau} = b_0 + \sigma W_{t \wedge \tau}$ is a stopped Brownian motion and hence a martingale. The expected exit time of $b_0 + \sigma W$ from $(-c^-, c^+)$ solves $\frac{\sigma^2}{2} u'' = -1$ with $u(-c^-) = u(c^+) = 0$, whose solution is $u(b_0) = (c^+ - b_0)(b_0 + c^-)/\sigma^2$. (b) With $\sigma = 0$ and $b > c^+$, $\dot{b} = -\kappa^+(b - c^+)$. The solution $b_t = c^+ + (b_0 - c^+)e^{-\kappa^+ t}$ stays above c^+ for all t , so the linear dynamics remain valid. The excess halves when $e^{-\kappa^+ t} = 1/2$. (c) For $b > c^+$, $\rho(b) = \kappa^+(b - c^+)/b = \kappa^+(1 - c^+/b)$, which is increasing in b and tends to κ^+ . On the band $\rho = 0$. The case $b < -c^-$ is symmetric. $\square$

Proof of Proposition 3. (a) Let $q > 0$ and $\sigma = 0$, so $\dot{b} = f(b) \equiv \beta q + \mu(b)$. The function f is continuous and strictly positive for $b \leq c^+$, strictly decreasing on (c^+, ∞) , and has a unique zero at $b^* = c^+ + \beta q/\kappa^+$. Since $f > 0$ below b^* and $f < 0$ above it, every trajectory converges monotonically to b^* . The case $q < 0$ is symmetric. If $q = 0$, f vanishes on the band and the argument of Proposition 2(b) applies outside it. (b) Differentiate b^* . (c) At the steady state $\dot{b} = 0$, so $\beta q = \lambda a^*$. $\square$

Proof of Corollary 1. Immediate from Proposition 3(a) applied to each market. $\square$

Proof of Proposition 6. Write $x_t = b_t - c$ and $a_t = K_t(x_t)_+ \geq 0$. The system is

$$\dot{x} = \beta q - \lambda a, \quad \dot{K} = \xi(\bar{K} - K) - \chi a.$$

The set $\{K > 0\}$ is forward invariant: at $K = 0$, $a = 0$ and $\dot{K} = \xi\bar{K} > 0$.

(a) A steady state requires $\lambda a = \beta q > 0$, so $x > 0$ and $a = Kx = \beta q/\lambda$. Then $\dot{K} = 0$ gives $K^* = \bar{K} - \chi\beta q/(\xi\lambda) = \bar{K}(1 - q/q_{crit})$, which is positive if and only if $q < q_{crit}$, and

$x^* = \beta q / (\lambda K^*)$. In the region $x > 0$ the Jacobian is

$$J = \begin{pmatrix} -\lambda K & -\lambda x \\ -\chi K & -\xi - \chi x \end{pmatrix}, \quad \text{tr } J < 0, \quad \det J = \lambda K(\xi + \chi x) - \lambda \chi K x = \lambda \xi K > 0,$$

so both eigenvalues have negative real parts at (x^*, K^*) , which establishes local asymptotic stability. Differentiating $b^* = c + \beta q / (\lambda \bar{K}(1 - q/q_{crit}))$ gives $\partial b^* / \partial q = \beta / (\lambda \bar{K}) \cdot [(1 - q/q_{crit}) + q/q_{crit}] / (1 - q/q_{crit})^2 = \beta / (\lambda \bar{K}(1 - q/q_{crit})^2)$, which is increasing and convex on $[0, q_{crit})$ and diverges as $q \uparrow q_{crit}$.

(b) Integrating the capacity equation over $[0, T]$ and using $K_T > 0$ and $K_t \geq 0$,

$$\chi \int_0^T a_t dt = \xi \bar{K} T - \xi \int_0^T K_t dt - (K_T - K_0) \leq \xi \bar{K} T + K_0.$$

Hence $\int_0^T a_t dt \leq \bar{a} T + K_0 / \chi$. Integrating the premium equation,

$$x_T = x_0 + \beta q T - \lambda \int_0^T a_t dt \geq x_0 - \frac{\lambda K_0}{\chi} + (\beta q - \lambda \bar{a}) T.$$

Dividing by T and letting $T \rightarrow \infty$ gives $\liminf x_T / T \geq \beta q - \lambda \bar{a} = \beta(q - q_{crit})$. If $q > q_{crit}$ this is positive, so no steady state can exist. $\square$

Proof of Proposition 4. After the shock, $b_0 = \delta$ and $\dot{b} = -\lambda a^*(b)$, with $\dot{p}^G = \lambda_G a^*(b)$. If $\delta \leq c^+$, $a^* = 0$ for all t and p^G does not move. If $\delta > c^+$, Proposition 2(b) gives $b_t - c^+ = (\delta - c^+)e^{-\lambda K^+ t}$. Since $\dot{p}^G = \lambda_G a^* = -\frac{\lambda_G}{\lambda} \dot{b}$,

$$p_h^G - p_0^G = \omega (b_0 - b_h) = \omega(\delta - c^+) (1 - e^{-\lambda K^+ h}).$$

Dividing by δ gives $S_h(\delta)$, and parts (a)–(c) follow by inspection. $\square$

Proof of Proposition 5. For a one-dimensional diffusion $db = \mu(b)dt + \sigma dW$ with Lipschitz drift, the speed measure has density $m(b) = \frac{2}{\sigma^2} \exp\left(\frac{2}{\sigma^2} \int_0^b \mu(x) dx\right)$. The process is positive recurrent, with stationary density proportional to m , whenever m is integrable and the scale function diverges at $\pm\infty$. Here $\int_0^b \mu = 0$ on the band, $-\frac{\kappa^+}{2}(b - c^+)^2$ for $b > c^+$, and $-\frac{\kappa^-}{2}(b + c^-)^2$ for $b < -c^-$. This gives (5), which is integrable. The scale density $1/m$ grows like a Gaussian reciprocal, so the scale function diverges at both ends. For the constant, use $\int_0^\infty e^{-\theta x^2} dx = \frac{1}{2} \sqrt{\pi/\theta}$ with $\theta = \kappa^\pm / \sigma^2$. Part (a) follows by integrating the plateau. For part (b), $\int_{-c^-}^{c^+} b db = \frac{(c^+)^2 - (c^-)^2}{2}$ and $\int_{c^+}^\infty b e^{-\theta(b - c^+)^2} db = c^+ \cdot \frac{1}{2} \sqrt{\pi/\theta} + \frac{1}{2\theta}$, with $\frac{1}{2\theta} = \frac{\sigma^2}{2\kappa^+}$; the lower tail is symmetric with the opposite sign. $\square$

Proof of Proposition 7. Since P is a projection and $P\mathbf{1} = 0$, $z = Pp$ satisfies $dz = -PF(p)dt + P\Sigma dW$, where $F_i(p) = \sum_j w_{ij}\phi_{c_{ij}}(p_i - p_j)$. By Itô's lemma, $V(z) = \frac{1}{2}\|z\|^2$ has generator

$$\mathcal{L}V = -z'F(p) + \frac{1}{2}\text{tr}(P\Sigma\Sigma'P),$$

where we used $z'P = z'$. By symmetry of w and oddness of ϕ ,

$$z'F(p) = \sum_i \sum_j w_{ij} z_i \phi_{c_{ij}}(p_i - p_j) = \sum_{i < j} w_{ij} (z_i - z_j) \phi_{c_{ij}}(p_i - p_j) = \sum_{i < j} w_{ij} (p_i - p_j) \phi_{c_{ij}}(p_i - p_j),$$

since $z_i - z_j = p_i - p_j$. For any x and $c > 0$, $x\phi_c(x) \geq \frac{1}{2}x^2 - \frac{1}{2}c^2$. This holds trivially for $|x| \leq c$; for $|x| > c$, $x\phi_c(x) = x^2 - c|x| \geq x^2 - \frac{1}{2}(x^2 + c^2)$. Therefore

$$z'F(p) \geq \frac{1}{2} \sum_{i < j} w_{ij} (p_i - p_j)^2 - \frac{1}{2} \sum_{i < j} w_{ij} c_{ij}^2 = \frac{1}{2} z' L_w z - \frac{1}{2} \sum_{i < j} w_{ij} c_{ij}^2.$$

Since $z \perp \mathbf{1}$, the Courant–Fischer theorem gives $z' L_w z \geq \lambda_2(L_w) \|z\|^2 = 2\lambda_2(L_w)V$. Thus

$$\mathcal{L}V \leq -\lambda_2(L_w)V + C, \quad C \equiv \frac{1}{2} \sum_{i < j} w_{ij} c_{ij}^2 + \frac{1}{2} \text{tr}(P\Sigma\Sigma'P).$$

By Dynkin's formula with a standard localization argument (the drift is Lipschitz, so all moments are finite), $\frac{d}{dt} \mathbb{E}V(z_t) \leq -\lambda_2 \mathbb{E}V(z_t) + C$. Grönwall's inequality then gives $\mathbb{E}V(z_t) \leq e^{-\lambda_2 t} \mathbb{E}V(z_0) + C/\lambda_2$, which yields the bound. If the network is disconnected, take two components A and B . The difference between their average prices has no drift, because arbitrage within each component preserves its average. If the noise is non-degenerate across components, that difference is a Brownian motion with positive variance rate, so its second moment, and hence $\mathbb{E}V$, grows linearly. $\square$

Proof of Corollary 2. (a) The Laplacian of the star $K_{1,N-1}$ with weight w has eigenvalues $0, w$ (multiplicity $N-2$) and Nw , so $\lambda_2 = w$. For any non-complete graph, Fiedler (1973) shows that algebraic connectivity is bounded above by vertex connectivity. Every tree on $N \geq 3$ vertices has vertex connectivity one, so $\lambda_2 \leq w$ for every spanning tree with weight w , and the star attains the bound. (b) Deleting the hub removes all edges. Deleting a leaf leaves a star on $N-1$ vertices, whose algebraic connectivity is again w for $N-1 \geq 3$ (for $N-1=2$ it is $2w$). Unbounded dispersion follows from the last part of Proposition 7. (c) The complete graph K_N with weight w has Laplacian $w(NI - \mathbf{1}\mathbf{1}')$, with eigenvalues 0 and Nw (multiplicity $N-1$). Removing a vertex leaves K_{N-1} . $\square$

B Further theoretical results

B.1 The stationary distribution of the premium

When order flow has mean zero and is short-lived relative to arbitrage, its effect is absorbed into the diffusion term. The premium then has an explicit stationary law.

Proposition 5 (Stationary distribution). *Let $q \equiv 0$. The premium process (3) is positive recurrent with unique stationary density*

$$\pi(b) = \frac{1}{Z} \begin{cases} \exp\left[-\frac{\lambda K^+}{\sigma^2}(b - c^+)^2\right], & b > c^+, \\ 1, & -c^- \leq b \leq c^+, \\ \exp\left[-\frac{\lambda K^-}{\sigma^2}(b + c^-)^2\right], & b < -c^-, \end{cases} \quad (5)$$

with normalizing constant

$$Z = c^+ + c^- + s^+ + s^-, \quad s^\pm \equiv \frac{1}{2} \sqrt{\frac{\pi \sigma^2}{\lambda K^\pm}}.$$

Consequently:

- (a) The stationary probability that the premium lies inside the band is $(c^+ + c^-)/Z$.
- (b) The stationary mean premium is

$$\mathbb{E}_\pi[b] = \frac{1}{Z} \left[\frac{(c^+)^2 - (c^-)^2}{2} + c^+ s^+ - c^- s^- + \frac{\sigma^2}{2\lambda K^+} - \frac{\sigma^2}{2\lambda K^-} \right],$$

which is non-zero whenever frictions or capacities are asymmetric, even though order flow has mean zero.

The density in (5) has a distinctive shape: a flat plateau whose width is the band, joined continuously to half-Gaussian tails whose precision $\lambda K^\pm/\sigma^2$ measures capacity relative to noise. Figure 7 compares it with a long simulated path. Part (b) cautions against reading a persistent average premium as excess demand. If it is harder to move dollars out of the country than in ($c^+ > c^-$ or $K^+ < K^-$), the premium is positive on average even when order flow is balanced.

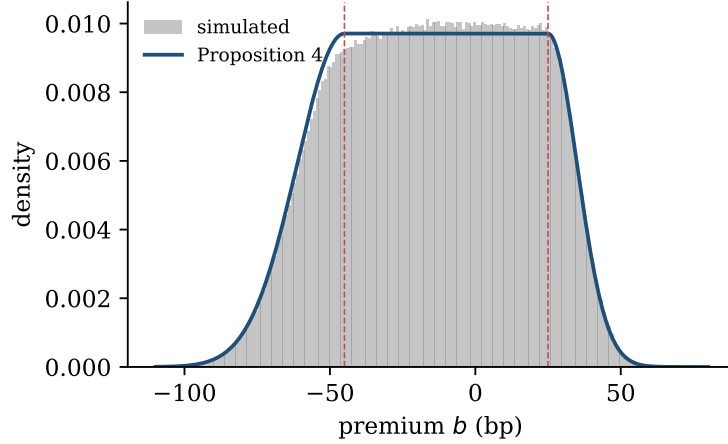


Figure 7: Stationary distribution of the premium. Asymmetric design with $c^+ = 25$, $c^- = 45$ bp, $\lambda K^+ = 0.08$, $\lambda K^- = 0.03$ per minute and $\sigma = 4$ bp/ $\sqrt{\text{min}}$. The histogram is a simulated path of 2×10^6 minutes; the solid line is the closed form (5). Dashed lines mark the band. The theoretical probability of being inside the band is 0.680, against 0.688 in the simulation (the small gap reflects Euler discretization at the boundary).

B.2 Capacity depletion and the arbitrage throughput limit

Capacity is depletable. Arbitrage ties up capital in transit, exhausts withdrawal limits and banking lines, and accumulates inventory that must be unwound. We now let capacity respond to arbitrage activity. Focus on the region $b > c^+$ and drop direction superscripts:

$$dK_t = [\xi(\bar{K} - K_t) - \chi |a_t^*|]dt, \quad a_t^* = K_t(b_t - c)_+, \quad (6)$$

where $\bar{K}$ is normal-times capacity, ξ is the rate at which capacity replenishes (capital is redeployed, limits reset), and χ is the rate at which arbitrage activity consumes it.

Proposition 6 (Throughput limit and convex impact). *Let $\sigma = 0$, $q_t \equiv q > 0$, and let (b_t, K_t) follow (3) and (6) with $K_0 > 0$. Define the arbitrage throughput limit $\bar{a} \equiv \xi \bar{K} / \chi$ and the critical order flow*

$$q_{crit} \equiv \frac{\lambda \bar{a}}{\beta} = \frac{\xi \lambda \bar{K}}{\chi \beta}.$$

(a) *If $q < q_{crit}$, there is a unique steady state,*

$$K^* = \bar{K} \left(1 - \frac{q}{q_{crit}}\right), \quad b^* = c + \frac{\beta q}{\lambda \bar{K} (1 - q/q_{crit})},$$

and it is locally asymptotically stable. The long-run price impact

$$\frac{\partial b^*}{\partial q} = \frac{\beta}{\lambda \bar{K} (1 - q/q_{crit})^2}$$

is increasing and convex in q and diverges as $q \uparrow q_{crit}$.

(b) If $q > q_{crit}$, no steady state exists, and for any initial condition

$$\liminf_{t \rightarrow \infty} \frac{b_t}{t} \geq \beta (q - q_{crit}) > 0.$$

Proposition 6 identifies the binding constraint in stressed markets. The binding constraint is the flow $\bar{a} = \xi \bar{K} / \chi$, the rate at which the arbitrage sector can sustainably process trades. Demand above this throughput limit cannot be absorbed at any premium, and the premium drifts away. This is the mechanism behind sudden de-pegs of local stablecoin prices during currency crises and capital-control tightening. Demand surges, arbitrage turns on, capacity is consumed, and future corrections slow down, which lets the premium keep widening. Figure 8 illustrates both regimes. In the illustrative calibration, depletion doubles the peak premium and raises the half-life of the subsequent correction from 14 minutes to more than six hours.

Remark 3 (A fluid-queue interpretation). Let $I_t \equiv \bar{K} - K_t$ be the balance-sheet capacity tied up in trades that have been executed but not yet settled. Equation (6) becomes

$$\dot{I}_t = \chi |a_t| - \xi I_t,$$

a fluid queue in which arbitrage trades arrive at rate $\chi |a_t|$ and settlement jobs, such as bank transfers and exchange withdrawals, are completed at rate ξ per unit in process. In steady state the work in process is $I^* = \chi a / \xi$, and it cannot exceed total capacity $\bar{K}$, so throughput is bounded by $\bar{a} = \xi \bar{K} / \chi$. Market clearing, $\lambda a = \beta q$, then gives $q_{crit} = \lambda \xi \bar{K} / (\chi \beta)$. The throughput limit of Proposition 6 is the capacity of the settlement queue, and the divergence of price impact as $q \uparrow q_{crit}$ has a heavy-traffic interpretation: as utilization q/q_{crit} approaches one, free capacity $K^* = \bar{K}(1 - q/q_{crit})$ vanishes and the premium needed to clear the market explodes. Settlement speed ξ , which depends on banking hours and withdrawal holds, enters the limit one-for-one.

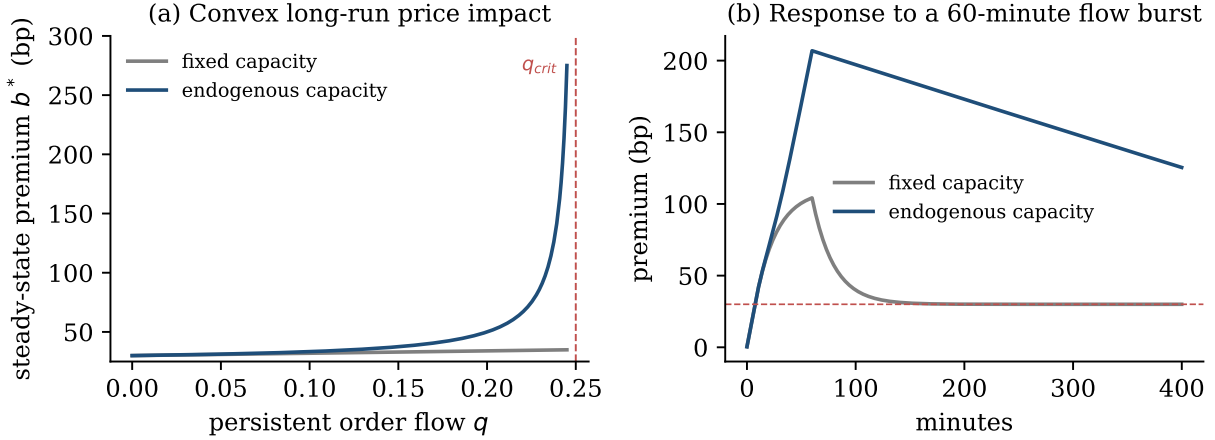


Figure 8: Capacity depletion. Parameters: $\beta = \lambda = 1$, $\bar{K} = 0.05$, $\xi = 0.01$, $\chi = 0.002$, $c = 30$ bp, so $q_{crit} = 0.25$. Panel (a): steady-state premium against persistent order flow, with fixed capacity (grey) and endogenous capacity (Proposition 6). Panel (b): response to order flow $q = 4$ for 60 minutes. With fixed capacity the post-burst half-life of the excess premium is 13.9 minutes, equal to $\ln 2 / (\lambda \bar{K})$; with endogenous capacity it is 370 minutes.

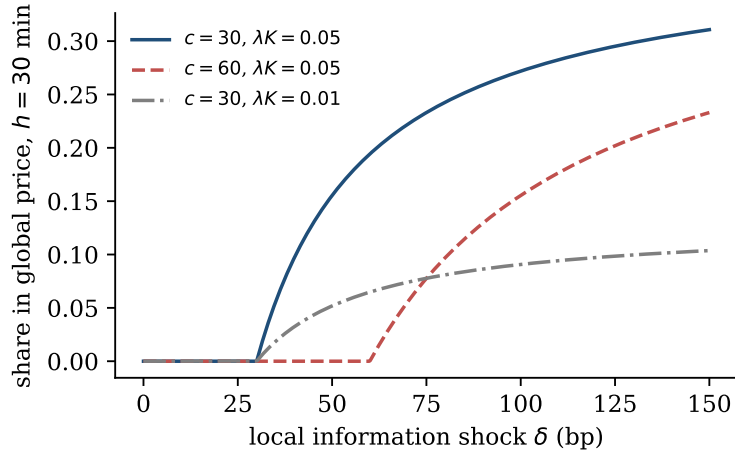


Figure 9: Transmission of local information to the global price. Fraction of a local information shock of size δ incorporated into the global price within $h = 30$ minutes (Proposition 4), with $\omega = 0.5$, for different frictions c and capacities λK .

B.3 Illustrations

B.4 Networks of venues

Stablecoins trade on many venues at once: several local exchanges, peer-to-peer platforms, and global order books. We now let $N \geq 3$ venues with log prices $p_t = (p_{1t}, \dots, p_{Nt})'$ be connected by pairwise threshold arbitrage. Let $w_{ij} = w_{ji} \geq 0$ be the capacity-scaled arbitrage intensity on link $\{i, j\}$, and $c_{ij} = c_{ji} > 0$ its friction. Let $\phi_c(x) \equiv \text{sign}(x) (|x| - c)_+$

denote the soft-threshold function. Prices follow

$$dp_{it} = - \sum_{j \neq i} w_{ij} \phi_{c_{ij}}(p_{it} - p_{jt}) dt + (\Sigma dW_t)_i, \quad (7)$$

where Σ is an $N \times N$ volatility matrix that also absorbs mean-zero order-flow shocks. Let L_w be the weighted graph Laplacian of (w_{ij}) , and let $\lambda_2(L_w)$ be its second-smallest eigenvalue, the *algebraic connectivity* of the arbitrage network (Fiedler, 1973). The network is connected if and only if $\lambda_2(L_w) > 0$. Cross-venue disagreement is measured by $z_t = Pp_t$ with $P = I - \frac{1}{N}\mathbf{1}\mathbf{1}'$ and $V(z) = \frac{1}{2}\|z\|^2$.

Proposition 7 (Dispersion and connectivity). *If the arbitrage network is connected, then*

$$\limsup_{t \rightarrow \infty} \mathbb{E}[V(z_t)] \leq \frac{\frac{1}{2} \sum_{i < j} w_{ij} c_{ij}^2 + \frac{1}{2} \text{tr}(P\Sigma\Sigma'P)}{\lambda_2(L_w)}.$$

If the network is disconnected, $\mathbb{E}[V(z_t)]$ grows linearly in t whenever the noise is non-degenerate across components.

The bound has two terms in the numerator. The first is the dispersion that frictions tolerate inside the bands; the second is the dispersion generated by noise. The denominator is the network's ability to pull prices together. Long-run dispersion therefore depends on capacity only through λ_2 , which depends on the *topology* of the network as well as on the level of capacity. Two corollaries follow.

Corollary 2 (Centralized arbitrage: efficient and fragile). *Suppose links are costly, so that the arbitrage network is a spanning tree, and each link has the same intensity w .*

- (a) *The star network, in which a single hub venue is linked to every other venue, attains $\lambda_2(L_w) = w$. This is the maximum over all spanning trees, so the hub structure minimizes the dispersion bound among minimal networks.*
- (b) *If the hub fails, all links vanish, the remaining $N - 1$ venues are mutually disconnected, and dispersion grows without bound. Removing any single non-hub venue leaves $\lambda_2 = w$ unchanged.*
- (c) *By contrast, in the complete network with intensity w on every link, $\lambda_2 = Nw$, and removing any venue leaves $\lambda_2 = (N - 1)w$.*

Corollary 2 formalizes a trade-off. When maintaining an arbitrage link is costly (accounts, banking relationships and transfer rails must be set up for each pair of venues), arbitrage naturally concentrates in a hub. In crypto the hub is typically the largest global exchange

or a dominant market maker (Hu and Zhang, 2023). The hub maximizes integration in normal times, but its failure disconnects the market. This mirrors, across secondary-market venues, the primary-market trade-off documented by Ma et al. (2025), and it predicts that the dispersion of local stablecoin prices should spike when a hub venue suspends withdrawals or a hub arbitrageur exits, by much more than when a peripheral venue does.

C A regulatory event: MASAK General Communiqué No. 29

General Communiqué No. 29 of the Financial Crimes Investigation Board (MASAK), published in the Official Gazette of 28 June 2025 and effective that day (MASAK, 2025), introduced holding periods of 48 hours for crypto-asset withdrawals and 72 hours for a customer’s first withdrawal, and limits on stablecoin withdrawals of USD 3,000 per day and USD 50,000 per month. Platforms may exempt transfers verified to be for liquidity provision, market making or inter-market arbitrage, subject to review and board approval. The rule constrains moving stablecoins out of Turkish platforms, the leg of the arbitrage that closes a discount.

Table 13 compares the lira market in the six months on either side of the communiqué, excluding the 19 March – 30 April 2025 political shock. Effective capacity is stable in levels on both sides of the band; the sensitivity of the premium to order flow falls from 0.47 to 0.32, and the average premium from 7 bp to zero. Table 14 adds USDT against the Mexican peso on the same exchange as a control market, measured against the interbank USD/MXN rate with the identical procedure. The declines in the premium and in order-flow impact are shared with, and larger in, the control market, so the cross-market comparison does not support attributing them to the regulation. The only differential movement is on the restricted leg: relative to the peso market, discount-side effective capacity is lower by 0.019 (s.e. 0.0075) after the communiqué, a relative divergence that reflects faster discount-side arbitrage in the control market while the lira market stayed flat. This pattern is consistent with the side targeted by the regulation; with one control market and one event, it is not offered as causal evidence.

D Monte Carlo evidence on the estimator

This section shows that the estimator recovers the structural parameters at realistic sample sizes and that linear models are misleading when the data are generated by threshold arbitrage. Units are basis points and minutes. The baseline design has $c^\pm = 30$ bp, $\kappa^\pm = 0.05$

Table 13: The lira market around MASAK General Communiqué No. 29 (28 June 2025)

	Before 28 Dec 2024 – 27 Jun 2025	After 28 Jun – 27 Dec 2025
κ^+	0.0060*** (0.0010)	0.0087 $\Delta = 0.0028$ (0.0024)
κ^-	0.0099*** (0.0016)	0.0091 $\Delta = -0.0008$ (0.0023)
β	0.473*** (0.034)	0.322 $\Delta = -0.150^{***}$ (0.043)
Mean lower edge ℓ_m (bp)	-6.0	-11.0
Mean upper edge u_m (bp)	28.7	8.3
Mean premium (bp)	7.1	-0.2
5th / 95th percentile of premium (bp)	-26 / 41	-29 / 29
N	22,961	30,269

Notes: Specification (4) with all slopes interacted with an indicator for the period after 28 June 2025; “ Δ ” is the interaction coefficient. The pre-period excludes 19 March – 30 April 2025. Band edges are averages of the monthly estimates. Month fixed effects; standard errors clustered by day.

Table 14: Difference-in-differences against USDT/MXN

	Control (MXN): change after 28 Jun 2025	Treated (TRY): change after 28 Jun 2025	Difference-in- differences
κ^+	0.0083 (0.0066)	0.0029	-0.0054 (0.0078)
κ^-	0.0181** (0.0071)	-0.0006	-0.0186** (0.0075)
β	-1.956*** (0.155)	-0.164	1.792*** (0.155)
Mean premium (bp)	-16.6*** (4.9)	-7.4	9.2** (4.2)
N (five-minute intervals)	53,497	53,229	106,726

Notes: Stacked specification (4) for USDT/TRY (treated) and USDT/MXN (control), 28 December 2024 – 27 December 2025, excluding 19 March – 30 April 2025 in both markets. Each market’s premium is measured against its interbank USD rate (Dukascopy) with month-specific bands; order flow is standardized within market-month. All slopes are interacted with market and post-28-June indicators; market-by-month fixed effects. The mean-premium row uses daily averages with market fixed effects and week-clustered standard errors. Other standard errors are clustered by day.

per minute (half-life 14 minutes), $\beta = 1$, $\sigma = 4 \text{ bp}/\sqrt{\text{min}}$, and order flow following a standardized AR(1) with autocorrelation 0.9. The asymmetric design has $c^+ = 20$, $c^- = 50$, $\kappa^+ = 0.08$ and $\kappa^- = 0.02$. The low-capacity design sets $\kappa^\pm = 0.015$. Each replication has $T = 20,000$ observations, about two weeks of minute data, and there are 200 replications per design. Thresholds are profiled over a grid with 2.5 bp spacing.

Table 15 reports the results. The thresholds are estimated with an RMSE of 2–3 bp in

Table 15: Monte Carlo: recovery of structural parameters

	c^+	c^-	$\kappa^+ = \lambda K^+$	$\kappa^- = \lambda K^-$	β
<i>Symmetric design</i>					
True	30.0	30.0	0.050	0.050	1.000
Median estimate	30.0	30.0	0.050	0.051	0.999
RMSE	2.5	2.7	0.007	0.007	0.029
Linear ECM $\hat{\kappa}$			0.0147		0.964
<i>Asymmetric design</i>					
True	20.0	50.0	0.080	0.020	1.000
Median estimate	20.0	50.0	0.081	0.020	0.999
RMSE	2.0	6.8	0.011	0.004	0.028
Linear ECM $\hat{\kappa}$			0.0105		0.963
<i>Low capacity design</i>					
True	30.0	30.0	0.015	0.015	1.000
Median estimate	30.0	30.0	0.016	0.015	0.999
RMSE	8.5	8.9	0.004	0.004	0.028
Linear ECM $\hat{\kappa}$			0.0075		0.991

Notes: 200 replications of $T = 20,000$ one-minute observations per design. “Median estimate” and “RMSE” refer to the profiled least-squares estimator of (4). “Linear ECM” reports the median reversion coefficient (first four columns) and order-flow coefficient (last column) from $\Delta b_{t+1} = \alpha + \beta OFI_t - \kappa b_t + \varepsilon_{t+1}$.

the baseline. Capacity-scaled speeds and order-flow impact are essentially unbiased. Asymmetries in both the band and capacity are recovered. In the low-capacity design the premium spends more time far outside the band, so its edges are visited less sharply and the thresholds are estimated less precisely, though still without bias. The implied long-run impact multipliers β/κ^+ have median estimates of 19.8, 12.3 and 63.9 against true values of 20.0, 12.5 and 66.7. The capacity differences that Corollary 1 says are hidden in raw premia are therefore recoverable from the joint dynamics of premia and order flow.

The linear error-correction model performs poorly. Its reversion coefficient is 0.015 in the baseline, less than a third of the true speed outside the band, and it implies a half-life of 47 minutes instead of 14. In the asymmetric design it averages two very different speeds into a single number. Reading the linear coefficient as a measure of arbitrage capacity, and $\beta/\hat{\kappa}$ as long-run impact, would overstate the long-run price impact of order flow by a factor of about two (low-capacity design) to more than three (baseline).

Table 16 implements test T2 on a long simulated path. Reversion is absent inside the band: the estimated slopes are economically negligible (at most 0.003 in absolute value, and not positive), and the implied half-lives are infinite. It switches on at the band edge and strengthens with the size of the dislocation, with the half-life falling from 68 minutes for

Table 16: Reversion speed by size of dislocation (baseline design)

$ b_t $ (bp)	[0,10)	[10,20)	[20,30)	[30,45)	[45,60)	[60,90)
Reversion speed per minute	-0.0027	-0.0011	-0.0003	0.0101	0.0212	0.0273
Implied half-life (min)	∞	∞	∞	68	33	25
Share of observations	0.175	0.182	0.190	0.271	0.143	0.039

Notes: Simulated path of 5×10^5 minutes. Within each bin of $|b_t|$, the table reports the slope from regressing $-\Delta b_{t+1} \text{ sign}(b_t)$ on $|b_t|$ and $OFI_t \text{ sign}(b_t)$ without an intercept, and the implied half-life $\ln 2/\text{slope}$ (reported as ∞ when the slope is not positive). The band is $[-30, 30]$ bp.

dislocations just outside the band to 25 minutes for the largest ones. This pattern is the empirical signature of Proposition 2(c), and it is inconsistent with any linear model.